Semester Project: Intelligent Techniques for Processing Large and Structured Data

Starting from **Week 4**, students will work on a **project** that will be developed progressively during the semester.

The goal of the project is to design and implement a **data-driven application capable of processing and extracting knowledge from large structured datasets**. Throughout the semester, the concepts presented in lectures and laboratories—such as **data exploration, feature engineering, scalable processing techniques, and intelligent algorithms**—will be gradually applied to the selected project.

In addition to the implementation, each team must write a **technical report describing the project and the obtained results**.

The report will document the complete project and must include sections such as:

- introduction and problem description
- dataset description
- related work
- data processing pipeline
- feature engineering
- modelling approach
- evaluation results
- discussion and conclusions

A **report template will be provided during the semester**, and all teams must follow this template when preparing their final report.

Project Objective

The objective of the semester project is to build a **complete data processing and analytics pipeline** capable of handling **large datasets** and extracting useful insights, predictions, or patterns from the data.

Each team will:

- select a **project topic**
- identify a **large dataset**

- process the data efficiently
- apply analytical or intelligent techniques
- evaluate the obtained results

The project should simulate a **real-world large-scale data analysis scenario**, similar to those encountered in modern data science and AI systems.

Minimum Application Requirement

In addition to the data processing and analytical components, each project must include **a minimal web/desktop application** that allows interaction with the developed system.

The purpose of this requirement is to demonstrate how the developed data analysis or intelligent model can be **integrated into a simple application layer**.

The web application does **not need to be complex**, but it should allow users to interact with the system in a meaningful way.

Students are free to use any appropriate framework, such as:

- **Flask**
- **FastAPI**
- **Django**
- **Streamlit**
- **Dash**

The goal is to provide a **basic demonstration interface** for the developed system.

Project Structure

Each team will select a **project subject** from the list proposed during the laboratory or propose a topic of their own (subject to instructor approval)- ONLY if it is a really nice and complex subject- **maximum 2 student proposals will be allowed to be implemented.**

The project will evolve throughout the semester and will gradually incorporate the techniques studied during the course, including:

- understanding the problem domain
- acquiring and preparing large datasets
- exploring and analyzing data
- engineering useful features

- developing analytical or predictive models
- evaluating results and discussing system limitations

Team Organization

Projects can be completed:

- **individually**, or
- **in teams of up to 3 students**

Teams may include students from different master programs if applicable.

Each team must send the team name, team members and chosen topic to the laboratory instructor.

A PROJECT SUBJECT CAN BE CHOSEN BY MAXIMUM 2 TEAMS.

Dataset Requirements

The project must use large datasets in order to reflect realistic data processing challenges.

Large Data Requirement

Datasets should contain at least 200,000 records.

Recommended dataset size:

- 200k – 1M rows

Students may use larger datasets if they have sufficient computational resources.

Structured Data Requirement

Projects must work with structured datasets or datasets that can be converted into structured form.

Structured attributes may include:

- numerical features
- categorical variables
- timestamps
- identifiers (user IDs, product IDs, etc.)

If the initial dataset is semi-structured (e.g., JSON logs, text metadata), students must transform it into a structured dataset before analysis.

Efficient Data Processing

Since the course focuses on **processing large structured data**, students are encouraged to use efficient tools designed for large-scale data processing.

Possible tools include:

- **Polars**
- **DuckDB**
- **PySpark**
- **SQL-based processing systems**

Students are also encouraged to use efficient storage formats such as:

- **Parquet**
- **ORC**

The goal is to demonstrate **efficient data handling and scalable processing techniques**.

Cloud Usage

Students may use **cloud-based infrastructure** if their local machines are not sufficient to process the dataset efficiently.

Possible platforms include:

- **Google Cloud Platform**
- **Amazon Web Services (AWS)**
- **Microsoft Azure**
- **Google Colab**
- **Kaggle environments**

Cloud storage services and distributed processing systems may also be used when appropriate.

Code and Reproducibility

Each project must include a **GitHub repository** containing:

- project source code
- data processing scripts
- model implementation
- web application code
- documentation explaining how to reproduce the results

During week 5 you will receive access to Github classroom for each team (based on their team name).

The repository should follow **clear documentation practices** to ensure that the project is reproducible (have a README file).

Final Deliverables

At the end of the semester, each team must submit:

1. Source Code Repository

A GitHub repository containing:

- the implementation
- data processing pipeline
- analytical or machine learning models
- the web application
- documentation

2. Technical Report

A structured report written using the **template provided during the course**, describing the methodology, experiments, and results.

3. Final Presentation

Each team will present their project during the last laboratory of the semester, including:

- problem description
- dataset used
- methodology
- system demonstration

- main results and insights

Evaluation Criteria

Projects will be evaluated every week and each stage will contain a certain weight from the total grade of the project.

The project grade is 40% from the final grade of this course and you need a minimum 5 grade to join the exam session.

In the retake session the projects and report cannot be presented !

- Each week you will have assignments regarding the application and the technical report and you will be graded accordingly.
- Each week delay is penalized with 3 points. After 2 weeks the assignment is graded with 0.
- During the laboratory you will present the application and technical report but the final grade will be given at the end of the week, ONLY after the detailed evaluation of the application and report (graded by both course and laboratory teachers).